

(A)

Total object = 8 = length of chromosomes

0/1 Knapsack

0 $\rightarrow$ Not taken

1 $\rightarrow$ Taken

0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1
A	B	C	D	E	F	G	H

Max Weight = 12

W $\rightarrow$ Weight

R $\rightarrow$ Reward

'H' must be taken.

1. Create initial population:

$C_1 \rightarrow 10010001 \rightarrow W=11, R=67$

$C_2 \rightarrow 10100101 \rightarrow W=10, R=62$

$C_3 \rightarrow 00101001 \rightarrow W=12, R=32$

$C_4 \rightarrow 01110011 \rightarrow W=20, R=66$

2. Fitness function:

if ($W \leq 12$):

Fitness = R

else:

Fitness = 0

Parent Selection: (Natural Selection)

2 Fittest chromosomes are

C_1 and C_2

Fitness(C_1) = 67 [$W(C_1) \leq 12$]

Fitness(C_2) = 62 [$W(C_2) \leq 12$]

Fitness(C_3) = 32 [$W(C_3) \leq 12$]

Fitness(C_4) = 0 [$W(C_4) > 12$]

3. Cross over: (Single Point)

$$\begin{array}{l} C_1 \rightarrow [1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1] \\ C_2 \rightarrow [1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1] \end{array} \xrightarrow{\text{Cross over}} \begin{array}{l} [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1] \rightarrow C_5 \\ [1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1] \rightarrow C_6 \end{array}$$

offspring are,

$$C_5 \rightarrow 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \rightarrow W=7, R=52$$

$$C_6 \rightarrow 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \rightarrow W=14, R=77$$

$$\text{Fitness}(C_5) = 52 \quad [W(C_5) \leq 12]$$

$$\text{Fitness}(C_6) = 0 \quad [W(C_6) > 12]$$

4. Mutation:

For C_5 : Let's changing 2nd and 3rd gene for better reward

$$C_5 \rightarrow 1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1 \rightarrow W=12, R=67$$

$$\text{Fitness}(C_5) = 67 \quad [W(C_5) \leq 12]$$

For C_6 : Changing 3rd gene to solve overweight problem

$$C_6 \rightarrow 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \rightarrow W=11, R=67$$

$$\text{Fitness}(C_6) = 67 \quad [W(C_6) \leq 12]$$

Final offspring are: $C_5 \rightarrow 11100101$

and, $C_6 \rightarrow 10010001$

B

Length of chromosomes = 7

Must visit every city, but only once.

1. Population Creation:

$$C_1 \rightarrow A F B C D G E \rightarrow \text{Cost} = 49$$

$$C_2 \rightarrow A D C F B E G \rightarrow \text{Cost} = 45$$

$$C_3 \rightarrow A B F C E D G \rightarrow \text{Cost} = 57$$

$$C_4 \rightarrow C E G D A B F \rightarrow \text{Cost} = 53$$

2. Fitness function:

$$\text{Fitness} = -\text{Cost}$$

$$\text{Fitness}(C_1) = -49$$

$$\text{Fitness}(C_2) = -45$$

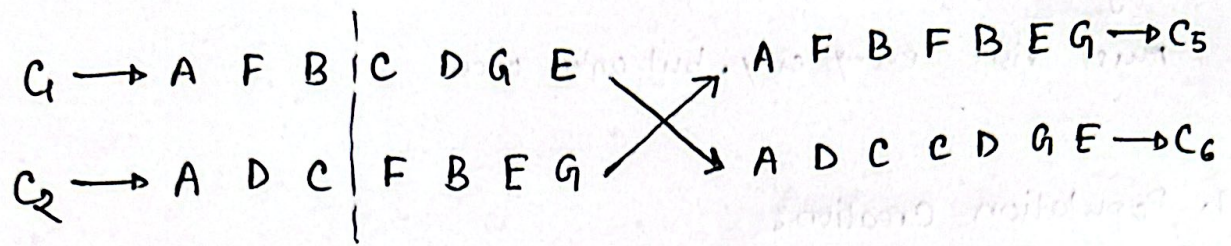
$$\text{Fitness}(C_3) = -57$$

$$\text{Fitness}(C_4) = -53$$

Parent Selection: (Natural Selection)

2 Fittest chromosomes are C_1 and C_2

3. Crossover (Single point on 3rd):



Here we can see, on the offspring C_5 and C_6 gene are repeated. However, we can visit a city only once, and every cities must be visited.

Therefore, offspring are not eligible.

4. Mutation:

Usual method of mutation is randomly replacing the gene.

In this case if we do that, it may create duplicate gene, remove any gene(city) or end up with invalid path.

Hence, usual method of mutation will not work here.

C

GBFS (Graph version)

→ A_{10}

× A_{10} $B_8 F_6$

× F_6 $B_8 G_{15} H_3$

× H_3 $B_8 G_{15} I_1$

× I_1 $B_8 G_{15}^F E_3 J_0 G_{15}^I$

✓ J_0 $B_8 G_{15}^F E_3^I G_{15}^I E_3^J$

Parent:

*	A			I	A	F	H	I	
A	B	C	D	E	F	G	H	I	J

Parent:

*	A			I	A	F	F	H	I
---	---	--	--	---	---	---	---	---	---

Visited: A F H I J

$P(J) = I$; $P(I) = H$; $P(H) = F$; $P(F) = A$

$P(A) = *$

∴ Path: $A \xrightarrow{3} F \xrightarrow{7} H \xrightarrow{2} I \xrightarrow{3} J$

Path cost = $3 + 7 + 2 + 3 = 15$

A* Search (Graph version)

→ A_{10}

× A_{10} $B_{14} F_9$

× F_9 $B_{14} G_9 H_{13}$

× G_9 $B_{14} H_{13} I_8$

× I_8 $B_{14} H_{13}^F E_{15} J_{10} H_{12}^I$

✓ J_{10} $B_{14} H_{13}^F E_{15} H_{12}^I$

Path: $A \xrightarrow{3} F \xrightarrow{1} G \xrightarrow{3} I \xrightarrow{3} J$

Total Path cost = 10

∴ The optimal path is $A \rightarrow F \rightarrow G \rightarrow I \rightarrow J$

With cost of 10.

Parent:

*	A			I	A	F	F	G	I
A	B	C	D	E	F	G	H	I	J

Visited: A F G I J

$f(B) = 6 + 8 = 14$

$f(F) = 3 + 6 = 9$

$f(G) = 3 + 1 + 5 = 9$

$F \rightarrow H$: $f(H) = 3 + 7 + 3 = 13$

$f(I) = 3 + 1 + 3 + 3 + 1 = 11$

$f(E) = 3 + 1 + 3 + 5 + 3 = 15$

$f(J) = 3 + 1 + 3 + 3 + 0 = 10$

$I \rightarrow H$: $f(H) = 3 + 1 + 3 + 2 + 3 = 12$

D

GBFS (Tree Search)

→ S_7

X S_7 $A_9 D_5$

X D_5 $A_9 B_4 E_3$

X E_3 $A_9 B_4 G_0$

✓ G_0 $A_9 B_4$

Parent:

*	S	D		S	D		E
	S	A	B	C	D	E	F

Path: $S \xrightarrow{2} D \xrightarrow{4} E \xrightarrow{3} G$

Total Path cost = 9

A* Search (Tree Search)

→ S_7

X S_7 $A_{12} D_7$

X D_7 $A_{12} B_7 E_9$

X B_7 $A_{12} E_9 C_7 E_7$

X C_7 $A_{12} E_9 E_7 G_9$

X E_7 $A_{12} E_9 G_9 G_7$

G_7 $A_{12} E_9 G_9$

Parent:

*	S	D		S	^{B_{E_7}} D_{E_9}		^{E_{G_7}} C_{G_0}
	S	A	B	C	D	E	F

$$f(A) = 2 + 9 = 11$$

$$f(D) = 2 + 5 = 7$$

$$f(B) = 2 + 1 + 4 = 7$$

$$D \rightarrow E: f(E) = 2 + 4 + 3 = 9$$

$$f(C) = 2 + 1 + 2 + 2 = 7$$

$$B \rightarrow E: f(E) = 2 + 1 + 1 + 3 = 7$$

$$C \rightarrow G: f(G) = 2 + 1 + 2 + 4 + 0 = 9$$

$$E \rightarrow G: f(G) = 2 + 1 + 1 + 3 + 0 = 7$$

Path: $S \xrightarrow{2} D \xrightarrow{1} B \xrightarrow{1} E \xrightarrow{3} G$

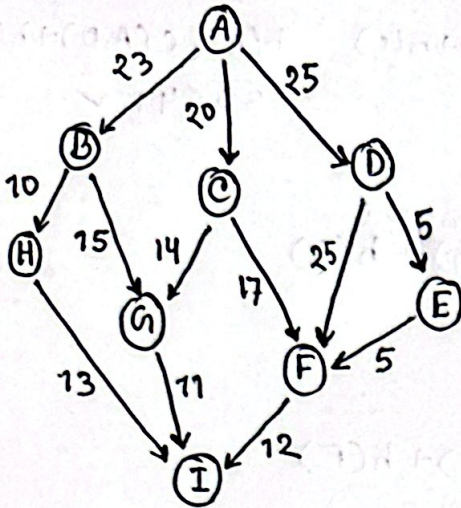
Total Path cost = 7

∴ The optimal path is $S \rightarrow D \rightarrow B \rightarrow E \rightarrow G$

With path cost of 7.

E

Node 'n' to goal node



Node	Min	Max
A	ACGI=45	ADFI=62
B	BHI=23	BGI=26
C	CGI=25	CEI=29
D	DEFI=22	DFI=37
E	EFI=17	EFI=17
F	FI=12	FI=12
G	GI=11	GI=11
H	HI=13	HI=13
I	I=0	I=0

$$A \xrightarrow{23} B \xrightarrow{10} H \xrightarrow{13} I = 46$$

$$A \xrightarrow{23} B \xrightarrow{15} G \xrightarrow{11} I = 49$$

$$A \xrightarrow{20} C \xrightarrow{14} G \xrightarrow{11} I = 45$$

$$A \xrightarrow{20} C \xrightarrow{17} F \xrightarrow{12} I = 49$$

$$A \xrightarrow{20} D \xrightarrow{25} F \xrightarrow{12} I = 62$$

$$A \xrightarrow{20} D \xrightarrow{5} E \xrightarrow{5} F \xrightarrow{12} I = 47$$

$$B \xrightarrow{10} H \xrightarrow{13} I = 23$$

$$B \xrightarrow{15} G \xrightarrow{11} I = 26$$

$$C \xrightarrow{14} G \xrightarrow{11} I = 25$$

$$C \xrightarrow{17} F \xrightarrow{12} I = 29$$

$$D \xrightarrow{25} F \xrightarrow{12} I = 37$$

$$D \xrightarrow{5} E \xrightarrow{5} F \xrightarrow{12} I = 22$$

$$H \xrightarrow{13} I = 13$$

$$G \xrightarrow{11} I = 11$$

$$F \xrightarrow{12} I = 12$$

$$E \xrightarrow{5} F \xrightarrow{12} I = 17$$

For Consistency,

$$h(n) \leq c(n, n') + h(n')$$

$$h(H) \leq 13, \quad h(G) \leq 11, \quad h(F) \leq 12,$$

$$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$$

$$11 \quad \quad \quad 9 \quad \quad \quad 10$$

$$h(E) \leq 5 + h(F), \quad h(B) \leq 10 + h(H)$$

$$\downarrow \quad \quad \quad \downarrow$$

$$13 \leq 15 \quad \quad \quad h(B) \leq 21$$

$$\downarrow$$

$$19$$

$$h(C) \leq 14 + h(G), \quad h(D) \leq 5 + h(E)$$

$$h(C) \leq 23 \quad \quad \quad h(D) \leq 5 + 19$$

$$\downarrow \quad \quad \quad \downarrow$$

$$21 \quad \quad \quad 16$$

$$h(A) \leq 20 + h(C)$$

$$h(A) \leq 41$$

$$\downarrow$$

$$39$$

1 Checking for Consistency,

$$\frac{A}{h(A) \leq C(A,B) + h(B)} \\ 39 \leq 42 \checkmark$$

$$h(A) \leq C(A,C) + h(C) \\ 39 \leq 41 \checkmark$$

$$h(A) \leq C(A,D) + h(D) \\ 39 \leq 41 \checkmark$$

$$\frac{B}{h(B) \leq C(B,H) + h(H)} \\ 19 \leq 21 \checkmark$$

$$h(B) \leq C(B,G) + h(G) \\ 19 \leq 24 \checkmark$$

$$\frac{C}{h(C) \leq C(C,G) + h(G)} \\ 21 \leq 23 \checkmark$$

$$h(C) \leq C(C,F) + h(F) \\ 21 \leq 27 \checkmark$$

$$\frac{D}{h(D) \leq C(D,F) + h(F)} \\ 16 \leq 35 \checkmark$$

$$h(D) \leq C(D,E) + h(E) \\ 16 \leq 18 \checkmark$$

$$\frac{E}{h(E) \leq C(E,F) + h(F)} \\ 13 \leq 15 \checkmark$$

$$\frac{G}{h(G) \leq C(G,I) + h(I)} \\ 9 \leq 11 \checkmark$$

$$\frac{F}{h(F) \leq C(F,I) + h(I)} \\ 10 \leq 12 \checkmark$$

$$\frac{H}{h(H) \leq C(H,I) + h(I)} \\ 11 \leq 13 \checkmark$$

Consistent heuristics are:

Node	heuristic
A	39
B	19
C	21
D	16
E	13

Node	heuristic
F	10
G	9
H	11
I	0

Q.1: Checking for Admissibility.

All consistent heuristics are Admissible.

However, let's check.

$$h(A) \leq c(A)$$

$$39 \leq 45 \checkmark$$

$$h(B) \leq c(B)$$

$$19 \leq 23 \checkmark$$

$$h(C) \leq c(C)$$

$$21 \leq 25 \checkmark$$

$$h(D) \leq c(D)$$

$$16 \leq 22 \checkmark$$

$$h(E) \leq c(E)$$

$$13 \leq 17 \checkmark$$

$$h(F) \leq c(F)$$

$$10 \leq 12 \checkmark$$

$$h(G) \leq c(G)$$

$$9 \leq 11 \checkmark$$

$$h(H) \leq c(H)$$

$$11 \leq 13 \checkmark$$

$$h(I) \leq c(I)$$

$$0 \leq 0 \checkmark$$

Admissible heuristics are:

Node	heuristic
A	39
B	19
C	21
D	16
E	13
F	10
G	9
H	11
I	0

3

Let's make, $h(A) = 43$
 $h(B) = 22$

$$h(A) \leq c(A)$$

$$43 \leq 45 \checkmark$$

$$h(B) \leq c(B)$$

$$22 \leq 23 \checkmark$$

Admissible. $\checkmark$

$$h(A) \leq c(A, B) + h(B) \rightarrow h(C)$$

$$43 \leq 20 + 21$$

$$43 \leq 41 \quad \times$$

$$h(B) \leq c(B, H) + h(H)$$

$$22 \leq 10 + 11$$

$$22 \leq 21 \quad \times$$

Not consistent.

Admissible but not consistent heuristic are.

Node	Heuristic
A	43
B	22
C	21
D	16
E	13
F	10
G	9
H	11
I	0

Node	Heuristic
A	43
B	22
C	21
D	16
E	13
F	10
G	9
H	11
I	0